Bowen Yuan (Bourne)

Visa Status: Permanent Residence

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Academic Profile

Computational design and extended reality researcher specialising in immersive technologies, digital communication methods, and spatial computing systems for architectural and urban contexts. My research develops spatio-temporal authoring frameworks and adaptive mixed reality environments that enable dynamic engagement with spatial narratives, environmental data, and procedural knowledge. My work bridges Human-Computer Interaction (HCI), Extended Reality (XR), computational design, and AI-augmented systems.

Education

PhD in Computer Science

University of South Australia (Adelaide University), Apr 2022 – Expected May 2026

Thesis: Dependency-Aware Mixed Reality Guidance

Supervisors: Assoc. Prof. Gun Lee; Prof. Mark Billinghurst

Visiting Researcher (Diminished Reality)

City University of Hong Kong, May 2025 – Sep 2025

Master of Information Technology

University of Queensland, Feb 2018 – Dec 2019

Bachelor of Electrical and Electronic Engineering

University of Liverpool, UK, Sep 2015 – Jul 2017

Academic Appointments

Lecturer (Sessional)

University of South Australia | Adelaide & Xi'an, Feb 2024 – Jul 2024

• Delivered studio-based Design Thinking curriculum.

• Led transnational teaching delivery in Xi'an.

• Designed workshops focused on user-centred design and spatial interaction systems.

• Provided curriculum alignment and assessment moderation.

Sessional Academic (Tutor & Practical Supervisor)

University of South Australia, Apr 2022 – Present

Courses: Data Structures Essentials; Cloud and Concurrent Programming; Introduction to Programming (Python).

• Delivered tutorials and laboratory sessions (20+ students).

• Explained advanced algorithmic concepts using visual modelling and live coding.

• Marked assessments and examinations with detailed technical feedback.

Research Contributions

• Developed STAGE framework for hierarchical dependency-graph AR replay systems.

• Designed immersive digital communication systems for architectural engagement.

• Engineered LLM-integrated Unity-based virtual agent systems.

Industry Experience

Software Developer

ByAsia Food Import Ltd., Sydney, Aug 2020 – Aug 2021

• Designed Warehouse Management System (WMS).

• Developed React Native logistics tracking application.

• Managed SQL databases and enterprise data integration.

Technical Skills

Programming: C++, Python, C#, Java, JavaScript (ES6), HTML/CSS

Frameworks: Unity3D, MRTK, React Native, React

AI & Systems: LLM API Integration, Python Torch

Academic Service & Engagement

• Peer reviewer for XR / HCI venues.

• International research collaboration (City University of Hong Kong).

• Member, ACM and IEEE.

Publication List

1. *Yuan, B., Xiao, Z., Cho, H., Teo, T., Lee, G. A., Billinghurst, M*. "Dependency-Aware AR Guidance for Replaying Non-Linear Work, " In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems 2026 (Poster), Barcelona, Spain, 2026.
2. *Yuan, B., Xiao, Z., Cho, H., Teo, T., Lee, G. A., Billinghurst, M.*
"Location-Based Non-Linear Playback of AR Task Recording, " In XRMemory, held in conjunction with the IEEE Conference on Virtual Reality and 3D User Interfaces 2026, Daejeon, South Korea, 2026
3. *Lee, G. A., Yuan, B., Hart, J. D., et al*. "Facilitating Discussions in Immersive Virtual Meetings with Virtual Agents,". In Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces 2026 (Poster), Daejeon, South Korea, 2026.
4. *B. Yuan, H. Cho, T. Teo, G. A. Lee and M. Billinghurst*, "Focus-Aware Task Guidance: Adaptive AR Instruction Playback Via Gaze and Location Tracking," *2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, Daejeon, Korea, Republic of, 2025, pp. 954-964, doi: https://doi.org/10.1109/ISMAR67309.2025.00103
5. *Yuan, Bowen, et al.* "Augmented reality spatial guidance cues for flexible multi-objective task environments." *Proceedings of the 19th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry*. 2024. DOI: https://doi.org/10.1145/3703619.3706047
6. *Yuan, Bowen, et al.* "The Fusion Nexus: Exploring the Confluence of Virtual and Real Worlds through Biocognitive Audio-Verbal Interface in Immersive XR Environments." *SIGGRAPH Asia 2023 XR*. 2023. 1-2. DOI: https://doi.org/10.1145/3610549.3614594
7. *Arindam Dey, Amit Barde, Bowen Yuan, et al.* "Effects of interacting with facial expressions and controllers in different virtual environments on presence, usability, affect, and neurophysiological signals." *International Journal of Human-Computer Studies* 160 (2022): 102762. DOI: https://doi.org/10.1016/j.ijhcs.2021.102762
8. *H. Cho, Bowen Yuan, et al.,* "An Asynchronous Hybrid Cross Reality Collaborative System," *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, Sydney, Australia, 2023, pp. 70-73, doi: https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00022.
9. *H. Cho, Bowen Yuan, et al.*, "Time Travellers: An Asynchronous Cross Reality Collaborative System," *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, Sydney, Australia, 2023, pp. 848-853, doi: https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00186.